

APPENDIX

TABLE III
PARAMETERS USED FOR MEMBERSHIP FUNCTION DEFINITIONS FOR EACH FRUGALITY SCORE CALCULATION.

Scenario	Dimension	Modality	a	b	c	d
Theoretical example	Accuracy	Low	0	0	0	0.5
		Medium	0	0.5	0.5	1
		High	0.5	1	1	1
	Energy in training (J)	Low	0	0	0	23760000
		Medium	0	23760000	23760000	71280000
		High	23760000	71280000	71280000	71280000
Non-neural machine learning on MNIST	Accuracy	Low	0	0	0	0.5
		Medium	0	0.5	0.5	1
		High	0.5	1	1	1
	Energy in training (J)	Low	0	0	0	275
		Medium	0	275	275	1375
		High	210	1375	49500	49500
	Energy in testing (J)	Low	0	0	0	137.5
		Medium	0	137.5	137.5	275
		High	137.5	275	16500	16500
Deep learning on ImageNet	Accuracy	Low	0	0	0	0.5
		Medium	0	0.5	0.5	0.75
		High	0.5	0.75	1	1
	Energy in training (J)	Low	0	0	0	23760000
		Medium	0	23760000	23760000	71280000
		High	23760000	71280000	498960000	498960000
	Energy in testing (J)	Low	0	0	0	2750
		Medium	0	2750	2750	8250
		High	2750	8250	33000	33000
Deep learning on CIFAR100	Accuracy	Low	0	0	0	0.5
		Medium	0	0.5	0.5	1
		High	0.5	1	1	1
	Energy in training (J)	Low	0	0	0	82500
		Medium	0	82500	82500	990000
		High	82500	990000	4950000	4950000
	Energy in testing (J)	Low	0	0	0	8250
		Medium	0	8250	8250	16500
		High	8250	16500	33000	33000
Deep learning on MNIST from BUTTER-E	Accuracy	Low	0	0	0	0.5
		Medium	0	0.5	0.5	1
		High	0.5	1	1	1
	Energy in training (J)	Low	0	0	0	3240000
		Medium	0	3240000	3240000	9720000
		High	3240000	9720000	19440000	19440000